

Sturm-Liouville and Orthogonal Functions

These are notes for a discussion about linear algebra and analysis for the students of the Research Club.

Linear algebra

Let V be an $\mathbf{R}$ -vector space, and recall that a linear operator L on V is a linear map

$$L : V \longrightarrow V$$

An eigenvector of L is a vector $v_\lambda \in V$ such that

$$Lv_\lambda = \lambda v_\lambda$$

The number $\lambda \in \mathbf{R}$ is called the eigenvalue of L associated to the eigenvector v_λ .

Suppose V is equipped with an inner product

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbf{R}$$

The adjoint $L^\dagger$ of L with respect to the inner product is a linear map

$$L^\dagger : V \longrightarrow V$$

defined by the formula

$$\langle v, L^\dagger u \rangle = \langle Lv, u \rangle \quad \forall \quad u, v \in V$$

Exercise 1. Let $V = \mathbf{R}^n$ and take the inner product to be the Euclidean one. Let L be a matrix and find the entries of $L^\dagger$ in terms of those of L .

Since $L^\dagger$ and L are both operators on V , we can compare them. An operator is called self-adjoint if $L^\dagger = L$.

Exercise 2. Let v_λ and v_μ be eigenvectors of L and suppose L is self-adjoint. Compare the “matrix elements” $\langle v_\mu, Lv_\lambda \rangle$ and $\langle v_\mu, L^\dagger v_\lambda \rangle$ to conclude that v_λ is orthogonal to v_μ (with respect to the inner product) if $\mu \neq \lambda$.

Consider the special case in which $\dim V = 2$ with the Euclidean inner product, and let $v_1, v_2 \in V$ be linearly-independent vectors. Compute the Euclidean length $|v_1|$, and define the unit vector

$$e_1 = \frac{v_1}{|v_1|}$$

We will now make a new vector v_2 defined by

$$v_2 := v_2 - \langle v_2, e_1 \rangle e_1$$

It is orthogonal to e_1 since e_1 is a unit vector. Compute the length $|v_2|$ and define the unit vector

$$e_2 = \frac{v_2}{|v_2|}$$

By construction

$$\langle e_i, e_j \rangle = \delta_{ij}$$

so $V = \text{Span}\{e_i\}_{i=1}^2$ is spanned by these “ortho(gonal)-normal(ized)” vectors.

Exercise 3 (Gram-Schmidt). *Use mathematical induction to extend this result to a procedure applicable to any finite-dimensional inner-product space. This algorithm is called the Gram-Schmidt orthonormalization process.*

Together with the result on orthogonal eigenvectors for self-adjoint operators, we conclude that if we have a vector space equipped with such an operator, we can construct an orthonormal basis of eigenvectors for the vector space.

Function Spaces

Orthogonal Polynomials

Let $S \subset \mathbf{R}$ be a subset of the real line and define the space $L^2(S)$ of square-integrable functions on S :

$$L^2(S) = \left\{ f : S \rightarrow \mathbf{R} \left| \int_S |f(x)|^2 dx < \infty \right. \right\}$$

Note that $L^2(S)$ is an inner-product space over $\mathbf{R}$ with

Consider $S = [-1, 1]$. Note that any continuous function defined on such a set is automatically (square) integrable.

Exercise 4. *Consider the monomials x^n defined on S . Starting with $p_0 = 1$, show that $p_1 = x$ is orthogonal to it. Find p_2 so that it is a quadratic polynomial orthogonal to both p_0 and p_1 . Find p_3 so that it is a cubic polynomial orthogonal to p_i with $i = 0, 1$, and 2 .*

Exercise 5. *Consider the operator on $L^2(S)$. Show that it is self-adjoint with respect to the inner product [innerprod].*

Hint: Show that any operator of the form $\frac{d}{dx} \left(p(x) \frac{d}{dx} \right)$ where $p(x)$ is a polynomial is “manifestly” self-adjoint, provided p vanishes on the boundary of S . Then find $p(x)$ for the operator above.

By convention, the Legendre polynomials P_n are normalized by

$$\langle P_m, P_n \rangle := \int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$$

The Rodriguez formula gives them as

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

These polynomials are the eigenvectors of the operator L of [LegendreLaplacian].

Exercise 6. Check this on the first few Legendre polynomials p_i for $i = 0, \dots, 3$. What are the eigenvalues?

Laplace equation

The Legendre polynomials are famous because they are instrumental to the solution of the Laplace equation $\Delta \varphi = 0$ in three dimensions. In spherical coordinates, the three-dimensional Laplacian operator is

$$\Delta = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \cdot \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \cdot \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2}{\partial \varphi^2}$$

To solve a problem with spherical symmetry, the *Ansatz* is made that the solution is separable: $\varphi(r, \theta, \varphi) = R(r)\Theta(\theta)\Phi(\varphi)$. This allows the separation of the Laplace equation into three parts. It can be shown that the $\Phi = \text{constant}$ and that the rest of the equation separates into the sum of two parts with one depending only on r and the other only on θ . This means that the two must be equal to the same constant. The θ -dependent part is given by

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + c\Theta = 0.$$

(The separation constant is denoted by c .)

Exercise 7. Make the change of variables $x = \cos \theta$ to show that this is an eigenvalue equation for the Legendre operator [LegendreLaplacian].

An aside on Taylor series

The completeness of eigenfunctions of a self-adjoint operator on an infinite-dimensional vector space is striking, but it should not be entirely shocking to a student of calculus. Recall that

a real function f of a real variable x is analytic on a domain $D \subset \mathbf{R}$ if there is a convergent series on D that agrees with the function f on D :

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \quad \text{on } D.$$

Exercise 8. Find an expression for the coefficients $\{a_n\}_{n=0}^{\infty}$ in terms of f and its derivatives.

(Anti-)Periodic Functions

Let $S = (-\pi, \pi)$, and consider the operator

$$L = \frac{d^2}{dx^2}$$

on $L^2(S)$. Then

$$\int g \frac{d^2}{dx^2} f dx = g \frac{df}{dx} \Big|_{x=-\pi}^{\pi} - \int \frac{dg}{dx} \frac{df}{dx} dx = g \frac{df}{dx} \Big|_{x=-\pi}^{\pi} - \frac{dg}{dx} f \Big|_{x=-\pi}^{\pi} + \int \left(\frac{d^2}{dx^2} g \right) f dx$$

The boundary terms will vanish if we restrict to the subset of $L^2(S)$ of functions that vanish at $x = \pm\pi$ or have vanishing first derivatives at those points. Let's call the former $\mathcal{D}(S)$ and the latter $\mathcal{N}(S)$.

Exercise 9. Find the eigenvectors and eigenvalues of the self-adjoint operator L on $\mathcal{D}(S)$ and $\mathcal{N}(S)$. Show that they are orthogonal and normalize them.

If the eigenvectors $\{v_n(x)\}_{n \in \mathbf{Z}}$, say of a self-adjoint operator span the vector space on which that operator acts, then in the context of $L^2(S)$, that must mean that it is possible to decompose any function $f \in L^2(S)$ in a (possibly infinite) sum of these eigenvectors $f(x) = \sum_{n \in \mathbf{Z}} c_n v_n(x)$. Alternatively, for some series of real coefficients $a_n, b_n \in \mathbf{R} \forall n$. This form is called the Fourier series representation of the function $f(x)$. It gives an entire function in terms of a series $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ of real numbers called the Fourier coefficients.

Exercise 10. Assuming this decomposition of the function and using the orthogonality of the eigen-functions you found above, calculate the a and b coefficients.

Complex Hilbert Spaces Complex numbers are crucial for all life, so we

would like to extend everything in sight to the complex domain. For one thing, that means that the vector spaces above will now be over $\mathbf{C}$. It turns out that the nicest inner product, however, should be real, or "Hermitian" to be more precise: It is linear in the second slot, and anti-linear in the first. Explicitly,

$$\langle v, au \rangle = a \langle v, u \rangle \quad \text{and} \quad \langle av, u \rangle = a \langle v, u \rangle$$

Exercise 11. Let V be an n -dimensional inner product space over $\mathbf{C}$ with Hermitian form $\langle \cdot, \cdot \rangle$. Let $L : V \rightarrow V$ be a linear operator on V . Then L has a matrix representation. If L is self-adjoint with respect to the inner product, what is the equivalent condition on its matrix representation?

A self-adjoint operator on a Hermitian inner-product space is also called a Hermitian operator.

Exercise 12. Show that the eigenvalues of a Hermitian operator are all real. Show that eigenvectors with distinct eigenvalues are orthogonal.

There is a great amount of very awesome stuff to discuss about complex variables. Unfortunately, we will have to skip all of it here. One thing we will need however is Euler's formula:

$$e^{ix} = \cos(x) + i \sin(x)$$

This is important to us for the following reason:

Consider extending the real Fourier analysis above to complex values (range but domain). Then $f : \mathbf{R} \rightarrow \mathbf{C}$ will have complex coefficients in the analogue of its Taylor expansion. Thus, the coefficients a and b in [FourierReal] are now allowed to be complex. Mediate for a second on the conclusion that the complex analogue of that result is that for a complex-valued function f

Exercise 13. For $m, n \in \mathbf{Z}$, show that¹

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-imx} e^{inx} dx = \delta_{m,n} := \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$$

Use this formula to derive an expression for the Fourier coefficients $\{c_n\}_{n=0}^{\infty}$.

Continuous Groups and Harmonic analysis The complex Fourier series

[ComplexFourierSeries] can be expressed as an integral over a group. In this case, the group $G = \mathbf{Z}$ is discrete and the integral reduces to a sum, but in general, the group could be continuous and even non-abelian. This is clear already from the formula for the Fourier modes

$$c_n = \int_{\mathbf{R}} f(x) e^{-inx} dx$$

...

¹The symbol in the middle of the equation above is the Kronecker δ ; it is defined by the expression on the right.